

Varun C

2nd Year B.E. ECE Student | Embedded Systems & AI Developer

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TECHNICAL SKILLS

Hardware & Embedded: ESP32, NodeMCU, Arduino, ESP32-CAM, Sensor Interfacing (I2C/SPI/UART), Breadboard Prototyping

AI & Software: ChromaDB, LLM Integration (LiteLLM), OpenCV, MediaPipe, REST APIs, WebSockets

Quant & Finance: Fyers API, Angel One API, Algorithmic Strategy Backtesting, yfinance, pandas

Languages: C, C++, Embedded C, Python, JavaScript, Verilog

Web & Backend: FastAPI, Next.js, React, Vite, Supabase, Firebase, Capacitor (Android APK)

Tools: Git, GitHub, Arduino IDE, PlatformIO, Verilator, Icarus Verilog, Vite

KEY PROJECTS

Hostel Management System — [Supabase](#) · [Vite](#) · [Capacitor](#) · [Chart.js](#)

Full-featured hostel management system with 5 user roles (Student, Caretaker, Gate, Mess, Admin), QR-based outpass, mess management, real-time dashboards, and audit logging. Built as a mobile-first web app with Android APK via Capacitor.

- QR code outpass scanning + generation with html5-qrcode
- Auto-attendance scheduler with background service
- Role-based access control with approval workflow
- Chart.js dashboards for data visualization

Project Zero — Autonomous AI Agent — [Python](#) · [LiteLLM](#) · [ChromaDB](#) · [FastAPI](#)

Advanced autonomous AI agent framework with 4 specialist roles (Scripter, Architect, Heavy, Sentry) routed via Gatekeeper. Persistent ChromaDB vector memory across all conversations. Native OS command execution and file I/O.

- Multi-model orchestration (Groq LLaMA + Gemini via LiteLLM)
- Autonomous goal decomposition via /goal prefix with Director agent
- Web UI with SSE streaming and execution logs
- Multi-modal vision analysis via base64 image encoding

AI Interview Simulator — [Next.js](#) · [FastAPI](#) · [Firebase](#) · [Framer Motion](#)

AI-powered virtual interview platform with resume parsing, automatic question generation (JD + resume overlap), and candidate scoring. Features AI-proctored video interviews with real-time transcription and TTS.

- 5-question flow: 3 technical + 2 behavioral with Fit Score (0-100)
- Resume parsing and competency extraction
- Admin dashboard with scoring rubric management and leaderboard
- Glass-morphism UI with dark theme

ALS — Eye-Blink Assistive Communicator — [Python](#) · [OpenCV](#) · [MediaPipe](#) · [pyttsx3](#)

Computer-vision-based assistive communication for people with motor disabilities. Uses MediaPipe Face Landmarker for dual-eye tracking and EAR algorithm for blink pattern detection.

- Blink patterns: 2 blinks = food, 3 = water, 4+ = emergency/call doctor
- Real-time TTS output via background thread
- Dashboard with blink count, EAR value, and eye state visualization

TRADE 2026 — Algo Trading Platform — [FastAPI](#) · [React 19](#) · [WebSocket](#) · [Angel One](#)

Full-stack algorithmic trading platform with paper/live trading, real-time WebSocket updates, and backtesting engine. Integrates with Angel One broker API.

- EMA Crossover + Supertrend + RSI strategy with configurable parameters
- Multi-symbol support (SBIN, TATASTEEL, ITC, TATAMOTORS, INFY)

- Real-time P&L charts via Recharts with WebSocket streaming
- Daily max loss/profit halting and risk per trade management

Virtual UI — Hand Gesture Control — [Python](#) · [OpenCV](#) · [MediaPipe](#) · [pyautogui](#)

Hand gesture-controlled virtual interface controlling multiple applications via webcam. Tracks 21 hand landmarks for real-time gesture recognition.

- Virtual mouse via index finger tracking
- Air writing/drawing on virtual pad
- Gesture-controlled Snake game and calculator
- PowerPoint slide control via hand gestures

Digital Clock Vibe — [Verilog](#) · [Icarus Verilog](#) · [FPGA](#)

Advanced digital clock design in Verilog with multiple operating modes. Features cascading counters, BCD-to-7-segment decoding, and button edge detection.

- 4 modes: CLOCK, SET_TIME, SET_ALARM, STOPWATCH
- 12/24 hour format toggle with AM/PM indicator
- Alarm with 10-second ring duration
- Stopwatch with seconds and hundredths precision

Fyers API Algorithmic Trading Bots — [Python](#) · [Fyers API](#) · [yfinance](#) · [pandas](#)

Automated trading signal generators for Indian stocks using EMA crossover and RSI strategies. Integrates with Fyers v3 API for real-time market data.

- BUY signals: EMA 9 above EMA 21, RSI > 50, price above EMA 200
- SELL signals: EMA 9 below EMA 21, RSI < 50, price below EMA 200
- Multi-Timeframe Analysis (MTFA) backtesting
- Fyers OAuth-based token generation with local HTTP server

OTHER PROJECTS

WhatsApp Simulator (Flutter · Firebase · Agora)
 LMS 3D Motherboard Simulation (Three.js · GSAP)
 AR Hand Tracking (MediaPipe Tasks · Web Audio)
 Smart Energy Meter Research (IoT Papers)
 AUTO MIST System (IoT Presentation)

Suma Coffee Shop Manager (React · Firebase)
 Interactive Birthday Card (Web Audio API · Canvas)
 NIFTY Options MTFA Backtester (Python · yfinance)
 Study Trackers (HTML/CSS/JS · LocalStorage)
 Energy Motor Monitor (Python)

LEADERSHIP & INITIATIVES

Founder @ MOON / Pyrose

Student innovation collective focused on building custom hardware prototypes, web applications, and AI solutions for real-world campus and community utility. Architected tools including the Hostel Management System adopted by university operations.

EDUCATION

B.E. in Electronics and Communication Engineering

Anna University — University College of Engineering Ariyalur (UCE Ariyalur) | Expected 2029

2nd Year | Relevant Coursework: Circuit Analysis, Digital System Design, Microcontrollers & Embedded Systems, Data Structures & Algorithms, VLSI Design